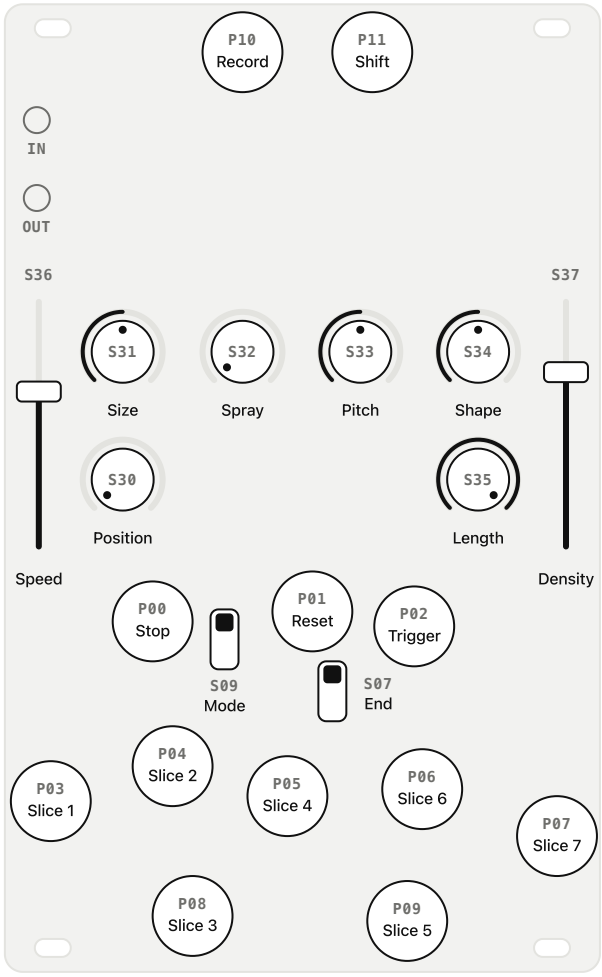
Simple Touch

# Clessidra

It's the instrument where you pour grains of sound from single drops into a dense cloud.

francescodimaggio.nl/clessidra

Synthux Design Residency 2026




Record a sound, then play it as grains: Speed through it, Density how they fall.

Two layers: each knob alone, and its twin with P11 held. Nothing jumps when they change.

Density at the bottom: no grains but the slices'. S09 at its centre: frozen.

## FADERS AND KNOBS

| | |
|---|---|
| S36 | **Speed**. Playback speed, 0 to 4×, the pitch kept. Frozen, it scrubs through the loop.<br>*P11*: **Volume**. Output level, off to 0 dB. |
| S37 | **Density**. Grains per second, 0.25 to 130, whatever their size. None at the bottom.<br>*P11*: **Random timing**. When grains start, from a regular pulse to random, as many per second. |
| S30 | **Position**. Where the loop starts. The playhead moves with it.<br>*P11*: **Filter**. Low-pass below the centre, off at the centre, high-pass above. |
| S35 | **Length**. How much of the rest the loop takes: at the top, to the end.<br>*P11*: **Reverb**. Reverb on the grains, from none to long tails. |
| S31 | **Size**. Grain length, 5 ms to 2 s, 100 ms at the centre.<br>*P11*: **Random size**. Each grain a different length, up to two octaves either way. |
| S32 | **Spray**. How far grains stray from the playhead, up to the whole loop, and how wide across the stereo field.<br>*P11*: **Reverse**. How many grains read the other way: none, some, all. The phrase keeps its direction, each grain turns. |
| S33 | **Pitch**. Transpose up to two octaves either way, with stops at musical intervals.<br>*P11*: **Random pitch**. Off at the centre. Above: each grain a different pitch, up to an octave either way. Below: in octaves, more grains an octave up or down. |
| S34 | **Shape**. Grain envelope: reversed attack, smooth at the centre, sharp attack.<br>*P11*: **Random shape**. Each grain a different envelope, from reversed to sharp. |

## PADS AND SWITCHES

| | |
|---|---|
| S09 | **Mode**. Forward, freeze or reverse. Frozen, Speed scrubs through the loop. |
| S07 | **End**. At the loop's end: once and no more, turned over to run back, or round again. Trigger pours it again. |
| P10 | **Record**. Tap to keep the last 8 s, hold to record up to 30 s. With Shift: listen. |
| P11 | **Shift**. Hold for each knob’s second function, shown in orange. |
| P00 | **Stop**. Stops as a tape machine, slowing to a halt. Touch again to start. |
| P01 | **Reset**. Second functions back to their defaults. |
| P02 | **Trigger**. Playback from the loop's start, with a grain at once: back on the beat. Stopped: starts at once. |
| P03–P09 | **Slices**. The loop in seven slices: its start, then the recording's strongest attacks inside it. Without attacks, as in a held note, even parts. Left to right on the board: P03, P04, P08, P05, P09, P06, P07. A touch: the playhead there, a grain at once, the beat from it. Held: it stays there. |

## TO PLAY

| | |
|---|---|
| Rhythm | Density low, Random timing (P11 + S37) down, Shape up, Size short. Size sets how long each hit lasts |
| Drops | Density near the bottom, Size short: single grains, now and then |
| Drums by hand | Density at the bottom, Shape up: tap the slices, several at once |
| Cloud | Density at the top, Spray up, Size long, Reverb (P11 + S35) up |
| Stretch | Speed at ¼ or ½, Density up. S09 down to hear it backwards |
| A moment held | S09 at the centre, Density up, a little Spray. Speed scrubs the point |
| One pour | S07 down: the loop plays once and stops. Trigger pours it again |
| Shimmer | Random pitch (P11 + S33) below the centre: octaves |
| Backwards syllables | Reverse (P11 + S32) at the top: the phrase in order, each grain backwards |

## LED

| | |
|---|---|
| Glow | Grains falling, dimmer as the sand runs out |

| | |
|---|---|
| `Flashing` | Recording |
| `Pulse` | At a stop: unison, speed 1×, 100 ms, Hann, filter off, random pitch off |
| `Steady` | P11 held: the twin last turned |